Ex.no:2	# EDA-DATA IMPORT AND EXPORT

Aim:

To inspect tourism dataset features, detect null values, and export data into CSV, Excel, and SQLite database formats.

Algorithm:

1. Import pandas and sqlite3 libraries.
2. Read tourism_dataset_200.csv using pd.read_csv().
3. Inspect data structure using .head(), .tail(), .info(), .describe(), and .isnull().sum().
4. Export DataFrame to .csv and .xlsx formats using .to_csv() and .to_excel().
5. Connect to tourism_database.db using sqlite3 and write data into an SQL table via .to_sql().
6. Close the database connection and end execution.

Program:

```
# Import required libraries
import pandas as pd
import sqlite3
# Load the Tourism dataset
df = pd.read_csv("tourism_dataset_200.csv")
# Display the complete dataset
print("===== TOURISM DATASET =====")
print(df)
# Display first five records
print("\n===== FIRST FIVE ROWS =====")
print(df.head())
# Display last five records
print("\n===== LAST FIVE ROWS =====")
print(df.tail())
# Display dataset information
print("\n===== DATASET INFORMATION =====")
print(df.info())
# Display statistical summary
print("\n===== STATISTICAL SUMMARY =====")
print(df.describe(include='all'))
# Display dataset shape
print("\nDataset Shape:", df.shape)
# Display column names
print("\nColumns:")
print(df.columns)
# Check missing values
print("\n===== MISSING VALUES =====")
print(df.isnull().sum())
# Export dataset to CSV
df.to_csv("tourism_dataset_export.csv", index=False)
print("CSV file exported successfully!")
```

```
# Export dataset to Excel

df.to_excel("tourism_dataset_export.xlsx", index=False)

print("Excel file exported successfully!")

# Create SQLite database

conn = sqlite3.connect("tourism_database.db")

# Export DataFrame to SQL table

df.to_sql(

"tourism_data",

conn,

if_exists="replace",

index=False

)

print("Dataset exported successfully to SQLite database!")

# Close database connection

conn.close()
```

Output:

```
===== TOURISM DATASET =====
    Tourism_ID     Destination       Country      Continent Tourism_Type  \
0         T001     Los Angeles           USA  North America         City
1         T002       Barcelona         Spain         Europe         City
2         T003           Paris        France         Europe     Wildlife
3         T004       Singapore     Singapore           Asia     Wildlife
4         T005           Tokyo         Japan           Asia     Cultural
..         ...             ...           ...            ...          ...
195       T196  Rio de Janeiro        Brazil  South America       Luxury
196       T197            Rome         Italy         Europe     Cultural
197       T198        Maldives      Maldives           Asia        Beach
198       T199       Cape Town  South Africa         Africa     Cultural
199       T200  Rio de Janeiro        Brazil  South America     Wildlife

    Best_Season  Average_Duration_Days  Estimated_Budget_USD  Rating  \
0        Winter                      3                   504     4.6
1        Winter                     11                  3756     3.5
2        Winter                     10                  1928     4.6
3        Spring                     14                   353     4.6
4        Spring                     14                  3057     3.7
..          ...                    ...                   ...     ...
195      Spring                      8                  1051     4.9
196      Spring                      8                  4412     4.1
197      Spring                      3                  3046     4.7
198      Summer                     13                  4142     4.2
199      Summer                      4                  4795     4.9
```

```
===== FIRST FIVE ROWS =====
  Tourism_ID  Destination    Country      Continent Tourism_Type Best_Season  \
0       T001  Los Angeles        USA  North America         City      Winter
1       T002    Barcelona      Spain         Europe         City      Winter
2       T003        Paris     France         Europe     Wildlife      Winter
3       T004    Singapore  Singapore           Asia     Wildlife      Spring
4       T005        Tokyo      Japan           Asia     Cultural      Spring

   Average_Duration_Days  Estimated_Budget_USD  Rating  Annual_Visitors  \
0                      3                   504     4.6           133393
1                     11                  3756     3.5            54123
2                     10                  1928     4.6           372696
3                     14                   353     4.6            88707
4                     14                  3057     3.7           204191

  Accommodation_Type Transportation
0              Hotel         Cruise
1          Apartment         Cruise
2           Homestay         Cruise
3             Hostel          Train
4             Hostel         Cruise

===== LAST FIVE ROWS =====
    Tourism_ID     Destination       Country      Continent Tourism_Type  \
195       T196  Rio de Janeiro        Brazil  South America       Luxury
196       T197            Rome         Italy         Europe     Cultural
197       T198        Maldives      Maldives           Asia        Beach
198       T199       Cape Town  South Africa         Africa     Cultural
199       T200  Rio de Janeiro        Brazil  South America     Wildlife
```

```
===== DATASET INFORMATION =====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   Tourism_ID             200 non-null    object 
 1   Destination            200 non-null    object 
 2   Country                200 non-null    object 
 3   Continent              200 non-null    object 
 4   Tourism_Type           200 non-null    object 
 5   Best_Season            200 non-null    object 
 6   Average_Duration_Days  200 non-null    int64  
 7   Estimated_Budget_USD   200 non-null    int64  
 8   Rating                 200 non-null    float64
 9   Annual_Visitors        200 non-null    int64  
 10  Accommodation_Type     200 non-null    object 
 11  Transportation         200 non-null    object 
dtypes: float64(1), int64(3), object(8)
memory usage: 18.9+ KB
None

===== STATISTICAL SUMMARY =====
       Tourism_ID Destination Country Continent Tourism_Type Best_Season  \
count         200         200     200       200          200         200   
unique        200          29      21         6            8           5   
top          T001   Marrakech   India      Asia     Heritage      Autumn   
freq            1          13      37        81           30          45   
mean          NaN         NaN     NaN       NaN          NaN         NaN   
std           NaN         NaN     NaN       NaN          NaN         NaN   
min           NaN         NaN     NaN       NaN          NaN         NaN   
25%           NaN         NaN     NaN       NaN          NaN         NaN   
```

Result: